

**Week 1-3:**

We submitted our proposal and RFA and had discussions with TAs about our initial ideas. Originally we wanted to go with a camera strapped to the user's forehead that could tell the user what chess piece to play. Ultimately, we changed to the idea for a chess robot out of concerns of our device being used for cheating and being too software focused. We also chose a magnetic arm because the machine shop already had a similar project in the past with it and also for scope, as an actual robotic arm or claw machine-like arm might be too complex.

**Week 4:**

We worked and finished the project proposal on 2/7 and 2/8. Previously, we talked with Jason about the basic designs of our project in the previous week on 1/30, where we decided on making a Pi-Hat with a Raspberry Pi and our PCB. We also talked with him about using the ESP32 S3 for our microcontroller, and I also decided that the original camera we wanted to use found here

([https://www.amazon.com/Charmed-Labs-Pixy2-Vision-Sensor/dp/B07D1CLYD2/ref=sr\\_1\\_1?crid=3FEJS3Y7TOCHT&keywords=pixy+smart+vision&qid=1706582260&srefix=pixy+smart+vision%2Caps%2C179&sr=8-1](https://www.amazon.com/Charmed-Labs-Pixy2-Vision-Sensor/dp/B07D1CLYD2/ref=sr_1_1?crid=3FEJS3Y7TOCHT&keywords=pixy+smart+vision&qid=1706582260&srefix=pixy+smart+vision%2Caps%2C179&sr=8-1)) was way overkill for the purposes of our project, specifically with the resolution used. These decisions we made are ultimately reflected in our project proposal.

**Week 5:**

February 18: We decided to utilize washers in our design instead of neodymium magnets. Additionally, I found some CAD designs for chess pieces here:  
<https://www.thingiverse.com/thing:378322>.

**Week 6:**

February 20: Yesterday, our focus was on conducting a cost analysis and finalizing the parts selection. Today, we visited the machine shop to acquire a button and talked about ensuring the magnet is flush with the board. I also obtained the measurements for the dimensions of the board (~18 by 19 inches), which will help us figure out the size of the chess pieces. I also got some washers from the machine shop so we can know what size to make the chess pieces.

February 21: We completed the design document and retrieved the chessboard from down at the machine shop. Additionally, we talked at office hours so we know now to use a MOSFET for the magnet switch. We got the 2.1mm barrel jack and calculated the cost of the chess pieces using UltiMaker Cura (pawns are 5 grams, knights are 9 grams, bishops and rooks are 8 grams, kings are 7 grams, and queens are 6 grams at 3 cents per gram for a total of \$6.18).

**Week 7:**

March 3: We successfully created three chess piece prototypes last Wednesday and completed the Kicad schematic to prepare for PCB review. During the PCB review, after discussing with TAs, we decided we are going to use a single buck converter for the 12V to 5V and then a linear regulator for the 5V to 3V. This is just so we are not losing efficiency for the 12V to 5V but we're

making it less complex for the 5V to 3V (rather than using a buck converter for that as well). Today, we focused on refining the schematic and selecting the correct stepper motor driver. We also fixed issues related to the magnet switch (we needed a schottky diode so that when the magnet turns off we have a place for the current to go to).

### Week 8:

March 4/5: I spent many hours fixing the PCB as Josh and Jose were busy with work and class. They helped out where they could when they didn't have class but it was difficult for all of us to work on the schematic at the same time. Essentially, 3/5 was spent turning the schematic into the actual PCB design, fixing issues with the parts (getting the right 2.1mm barrel jack footprint, getting the right linear regulator, getting the right MOSFET footprint, etc.), and trying to wire it in a way that was not cluttered on the PCB. We were late with the PCB, but we were able to get an order for the first round of PCB orders. Here are some of the parts that were not included from Jason's schematic (which we modeled ours after):

[DRV8434SPWPR Texas Instruments | Mouser](#)

[LM3940IT-3.3 Texas Instruments | Integrated Circuits \(ICs\) | DigiKey](#)

[PJ-002AH CUI Devices | Connectors, Interconnects | DigiKey](#)

This is the final schematic and PCB for round 1:



